

4.1 Designing Schemas

Is this schema good?

• Schema: students (id, name, major, birthday)

• Instance:

id	name	major	birthday
1	Bugs Bunny	CS	2004-11-06
2	Donald Duck	Bio	1997-02-01
3	Peter Pan	Econ	1998-10-01
4	Mickey Mouse	CS	1995-04-01

* If student has multiple majors?

↳ duplicate data (w/ name and birthday)

Anomalies

- Redundancy
- Update anomaly (when updating birthday)

* If student has no major?

↳ no row for that student (lose all info) or NULL in major.

Either way some storage isn't needed for major and birthday.

Problem: Storing Birthday with major.

• When a student has one major.

↳ Schema is good.

• When a student can have multiple or no majors

↳ The same schema is bad.

↳ Problems:

- ↳ Redundancy
- ↳ Loss of facts
- ↳ Update anomaly

• The goodness of a schema depends on the domain characteristics.

How do we consider a schema good?

• A good schema is called a normal form.

• There are various normal forms.

• We will discuss Boyce-Codd Normal Form (BCNF)

How do we transform a bad schema into a good one?

• That is, to normalize a schema to a normal form.

4.2 Functional Dependencies

Functional dependency

- A form of constraint, and therefore part of a schema. ↳ key, referential integrity
- A key factor for determining and organizing good schemas.

About Major and Birthday

- How many majors do you have?
 - None, one, or multiple.
 - $id \rightarrow major$? NO
(unique major given id?)

• How many birthdays do you have?

- one
- $id \rightarrow birthday$? YES

Functional Dependency

• Notation: $A_1, \dots, A_m \rightarrow B_1, \dots, B_n$

• We say: $A_1, \dots, A_m$ functionally determines $B_1, \dots, B_n$

• Meaning:

- If any tuples agree on $A_1, \dots, A_m$ values, then they must also agree on $B_1, \dots, B_n$.
- i.e. the mapping from $A_1, \dots, A_m$ to $B_1, \dots, B_n$ is functional (many-to-one).

Whether FDs hold is your knowledge of the domain.

- $id \rightarrow birthday$
- $id, course \rightarrow grade$

A "functionally determines" B

- Given a value of A, there exists at most one value of B -- no ambiguity.
- It does not mean: B can be computed from A by a formula.

FD: Your Domain Knowledge/Assumption.

- Understand domain and make assumption accordingly.
- $id \rightarrow major$
 - possibly true, if a student can have only one major.
- $id, course \rightarrow grade$
 - possibly false, if a student can take a course multiple times.

4.3 Keys

What is a Key?

- Key of a relation $R(A_1, \dots, A_n)$ is a set of attributes K that:
 - Functionally determines all attributes of R, $K \rightarrow A_1, \dots, A_n$, and
 - None of its subsets does
- Superkey
 - A set of attributes that contains a key.

4.4 Reasoning with FDs

Basic Rules: Armstrong's Axioms

- Reflexivity rule $id, name \rightarrow name$
 - If $B \subseteq A$, then $A \rightarrow B$
- Augmentation rule
 - If $A \rightarrow B$, then $AC \rightarrow BC$
- Transitivity rule
 - If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.

Deriving More Rules

- Splitting Rules
 - If $A \rightarrow BC$, then $A \rightarrow B$, and $A \rightarrow C$.
- Combining Rule
 - If $A \rightarrow B$, and $A \rightarrow C$, then $A \rightarrow BC$.

Given

- $drinker, bar, beer \rightarrow season$
- $drinker, bar \rightarrow beer$
- $bar, beer \rightarrow price$

Decide if $\{drinker, bar\}$ is a Superkey?

- 1) $drinker, bar \rightarrow drinker, bar$ (reflexivity)
- 2) $drinker, bar \rightarrow beer$ (given)
- 3) $drinker, bar \rightarrow season$

- $drinker, bar \rightarrow drinker, bar, beer$ (augmenting)
- $drinker, bar, beer \rightarrow season$ (given)
- $drinker, bar \rightarrow season$ (transitivity)

4) $drinker, bar \rightarrow price$

4.5 Attribute closures

What attributes can {drinker, bar} determine?

This is called the closure of {drinker, bar}.

Closure of Attributes

• Problem: What can these attributes determine?

- Given a set of attributes $\{A_1, \dots, A_n\}$ and a set of dependencies F .
- Find all attributes $B_1, \dots, B_m$ such that any relation that satisfies F also satisfies:

$$A_1, \dots, A_n \rightarrow B_1, \dots, B_m$$

- The closure of $\{A_1, \dots, A_n\}$ is $B_1, \dots, B_m$
i.e. $\{A_1, \dots, A_n\}^+ = \{B_1, \dots, B_m\}$

Finding Attribute closures

- Given a set of attributes $\{A_1, \dots, A_n\}$ and a set of dependencies F .
- $C = \{A_1, \dots, A_n\}$
- Repeat until C does not change:
 - If $X_1, \dots, X_m \rightarrow Y$ is in F , and $X_1, \dots, X_m$ are all in C , and Y not in C :
 - $C \leftarrow C \cup Y$

Normal Forms

(23)

Problematic schema

id, name, major, birthday

repeated name and birthday if a student has multiple majors.

A Better Design

students

id, name, birthday

Majors

id, major

- Redundancy?
- Update Anomaly?
- Delete Anomaly?

Schema Refinement: Desired Properties

- Minimize Redundancy
- Avoid info loss
- Preserve dependency
- Ensure good query performance.

First Normal Form

1NF: Each attribute contains only single atomic values.

Unnormalized set vs Normalized set.

Normal Forms

- Many normal forms were proposed, as more desired properties were discovered.
 - 2NF, 3NF, ~~4NF~~ BCNF, 4NF

Boyce-Codd Normal Form

id, name, major, birthday

What's the culprit?

- id $\rightarrow$ name, birthday
- id $\rightarrow$ major
- id determines some attributes A.
- id does not determine other attributes i.e. it is not a superkey.
- Thus, A will repeat.

• Lesson: An FD by non-key attributes can cause redundancy.

Boyce-Codd Normal Form

• A relation R is in BCNF if and only if:

Whenever there is a nontrivial FD for R,

$$A \rightarrow B$$

then A is a superkey for R.

• Whenever a set of attributes of R determines another attribute, it should determine all attributes of R.

• That is, no bad FDs!

BCNF decomposition

• Algorithm BCNF

• Input: Relation R, FDs F

• If (exists an FD $A \rightarrow B$ that violates the BCNF condition)

• Decompose $R(A, B, C)$ into $R_1(A, B)$ and $R_2(A, C)$

• Compute FDs for R_1 and R_2 as F_1 and F_2 .

• Return $BCNF(R_1, F_1) \cup BCNF(R_2, F_2)$

• Else
• Return R

4.8 Lossless Decomposition

29

A decomposition can be lossy

• Favourites (drinker, bar, beer)

drinker	bar	beer
Alex	John Bar	Sam Adams
Carissa	Green Bar	Bud Light
Alex	Purple Bar	Coors

• Decomposition

$R_1 =$		$R_2 =$	
drinker	bar	drinker	beer
Alex	John Bar	Alex	Sam Adams
Carissa	Green Bar	Carissa	Bud Light
Alex	Purple Bar	Alex	Coors

• Favourites = $R_1 \bowtie R_2$

• No! We get extra tuples:

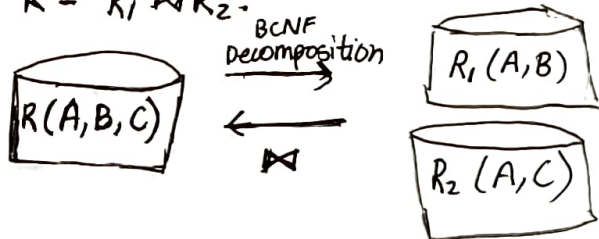
- (Alex, John Bar, Coors)
- (Alex, Purple Bar, Sam Adams)

BCNF is lossless decomposition

• That is, we reconstruct the original relation.

• If we decompose $R(A, B, C)$ into $R_1(A, B)$ and $R_2(A, C)$ due to $A \rightarrow B$, then

$$R = R_1 \bowtie R_2$$



4.9 Dependency-Preserving Decomposition

A decomposition may not preserve dependency.

- A schema S is dependency preserving if, for every FD f of it:
 - f can be checked in a table T in S , or
 - f can be implied by those FDs that can be checked in single tables.

BCNF May Not Preserve Dependency

- Yes, this is lossless.
- But it does not preserve dependency

4.10 3rd Normal Form

3NF: Making a compromise from BCNF

- A relation R is in 3NF if and only if:
Whenever there is a nontrivial FD for R ,
 $A \rightarrow B$
then A is a superkey for R , ~~or~~
or B is a prime attribute (i.e. member of a key) for R .

Why do we consider 3NF Acceptable?

- 3NF decomposition is both:
 - lossless decomposition
 - dependency preserving
- It removes "bad FDs" mostly
 - Except those involving key attributes
 - i.e., will not split a key in two relations.

4.11 Multivalued Dependencies

(25)

More than Functional Dependencies

- There are other kinds of dependencies than FDs.
 - Multivalued dependencies (MVD)
 - Inclusion dependencies (IND)
 - Join dependencies (JD)

Consider our Academic World

- We know $id \rightarrow birthday$,
 id determines birthday.
- But id can also determine major as a set of majors instead of a unique value.
- We call this determination a multivalued dependency:
 $id \twoheadrightarrow major$
 $\downarrow$
multidetermines

MVD is "Tuple Generating"

students ($id, name, major, gpa, hobby, level$)

suppose $id \twoheadrightarrow major, gpa$

given id you know major and gpa value groups.

Multivalued Dependency (MVD)

- Notation: $A_1, \dots, A_m \twoheadrightarrow B_1, \dots, B_n$
- We say: $A_1, \dots, A_m$ multidetermines $B_1, \dots, B_n$
- Meaning:
 - If two tuples agree on $A_1, \dots, A_m$ values, then swapping their $B_1, \dots, B_n$ values will result in two tuples that are also in the relation.
 - i.e., B depends only on A , and is independent of the remaining attributes.

FD: Special case of MVD

(26)

- FD is a special case of MVD.
- $id \rightarrow birthday \Rightarrow id \twoheadrightarrow birthday$
 $\downarrow$ $\downarrow$
1 value 1 or more values

FD / MVD Are Domain Knowledge